

OFFLINE KNOWLEDGE PREPARATION

Class labels
 $\times 47$ relations

RotatE
Top-M tails

AAKV
triple descriptions

CLAP text
cache

ONLINE SAMPLE-LEVEL INFERENCE

CLAP INITIAL RANKING

Input audio + candidate class text

Base scores s_{base} $\rightarrow$ Top- K classes

AUDIO-EVIDENCE MATCHING

Audio + cached AAKV text embeddings

Evidence scores u_{irt}

RELATION EXPERT RESULTS

Relation

Top-1 class

Top-two score gap Δ_r

r_1

A

Δ_1

r_2

A

Δ_2

r_3

B

Δ_3

$\vdots$

$\vdots$

$\vdots$

r_{47}

A

Δ_{47}

Top-1 votes $\rightarrow$ Consensus class: A

Relations voting for A first; sort each group by Δ_r (descending)

Select the first 5 relations

SELECTED EVIDENCE PER CLASS

Pool tails from the 5 relations

Exact-string dedup $\rightarrow A_k(\mathcal{U}_i^*)$

FINAL SCORES AND RANKING

$$s_{\text{final}}(c_i) = \alpha s_{\text{base}}(c_i)$$

$$+ (1 - \alpha) A_k(\mathcal{U}_i^*)$$

Rank all classes $\rightarrow$ predicted label